NOT MEASUREMENT SENSITIVE
MIL-P-85856/34
7 December 1988

MILITARY SPECIFICATION SHEET

# POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS ALUMINUM OXIDE, TYPE II, CLASS 1, COMPOSITION D

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

## Chemical Composition

| $SiO_2$ | $Na_2O$ | $Al_2O_3$ | Others (1) | |
|---|---|---|---|---|
| | | | Each | Total |
| 1.0 max. | 1.0 max. | 98.0 min. | - | 0.75 max. |

(1) Determination not required for routine acceptance.

## Particle Size/Distribution

| | |
|---|---|
| -270 Mesh (-53 µm) | 100.0 min. |
| -325 Mesh (-45 µm) | 99.0 min. |

## Sprayed Coating Properties

| | |
|---|---|
| Bond Strength | 45.0 MPa (2000 psi) min. |
| Hardness | Rockwell 15N, 80 min. |
| Microstructure | See paragraph 4.5.4.1.2<br>Porosity - 15% max.<br>Unmelted Particles - 5% max. |

AMSC N/A FSC 3439

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Deposit Efficiency

70% min for $N_2/H_2$
65% min for $A/H_2$

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

| | |
|---|---|
| Pratt & Whitney Aircraft | PWA 1310 |

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/34)



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